

DISSERTATION REPORT
ON
"COLLEGE ADMISSION ENQUIRY SYSTEM"

Submitted in partial fulfilment of the requirements for the award of the degree of
BACHELOR OF COMPUTER APPLICATIONS (BCA)
Semester VI

Submitted By

Name of Student: _____

Roll No.: _____

University Enrolment No.: _____

Under the Guidance of

Name of Supervisor: _____

Department of Computer Applications

B.J.S. Rampuria Jain College, Bikaner

Academic Session: 2026-27

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B.J.S. RAMPURIA JAIN COLLEGE, BIKANER
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CERTIFICATE

This is to certify that Mr./Ms. _____, Roll No. _____, Enrolment No. _____, student of BCA Semester VI, B.J.S. Rampuria Jain College, Bikaner, has successfully completed the dissertation project entitled "College Admission Enquiry System" under my guidance and supervision during the Academic Session 2026-27.

The work presented in this dissertation is based on the student's study, analysis, development and implementation carried out as part of the academic requirements of the BCA Semester VI programme.

To the best of my knowledge, the work submitted by the student is suitable for evaluation as a Semester VI Dissertation Project.

Date: _____

Place: Bikaner

Signature of Project Supervisor

Name: _____

DECLARATION

I, Mr./Ms. _____, student of BCA Semester VI, Roll No. _____, hereby declare that the dissertation project entitled "College Admission Enquiry System" submitted by me to B.J.S. Rampuria Jain College, Bikaner, during the Academic Session 2026-27, is my original academic work carried out under the guidance of _____.

I further declare that this project has been prepared for the purpose of academic evaluation and has not been submitted previously, either fully or partially, for the award of any other degree, diploma or certificate.

Wherever information, concepts, code, images, datasets or other material from external sources have been used, appropriate acknowledgement and references have been provided.

I understand that plagiarism, fabrication of results, submission of copied work or misrepresentation of another person's work may lead to rejection of the project and/or disciplinary action as per institutional rules.

Date: _____

Place: Bikaner

Signature of Student: _____

Name: _____

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Finally, I acknowledge all books, websites, software tools, libraries, datasets and other resources that helped me understand the concepts and complete this project successfully.

Name _____ of _____ Student: _____

Roll No.: _____

ABSTRACT

The project entitled "College Admission Enquiry System" focuses on digitizing and streamlining the student enquiry and admission tracking process. The primary objective of the project is to provide a centralized, user-friendly platform for prospective students to explore course offerings and submit enquiries, while equipping college administration with a robust backend to manage these leads efficiently. The project has been developed using Python Flask framework and makes use of MySQL for data storage, along with HTML/CSS and JavaScript for a responsive front-end design.

The proposed system provides facilities such as dynamic course and faculty listing, a secure online enquiry submission form with server-side validation, and a comprehensive admin dashboard for tracking, filtering, and updating enquiry statuses. The methodology adopted for the project includes a study of the existing manual system, identification of requirements, system design, implementation using modern web development practices, testing, and evaluation of the developed solution.

The developed project was tested using appropriate test cases to verify its major functionalities. The results indicate that the proposed system can be used successfully to automate the enquiry management process, reducing manual errors and improving communication efficiency. The project provided practical exposure to concepts related to web development, database management, and backend programming, and helped in understanding the process of converting theoretical computer application concepts into a practical solution.

Chapter 1 - Introduction

1.1 Background of the Project

The College Admission Enquiry System is a web-based application designed to streamline the process of handling student enquiries and managing admission-related information. Traditionally, colleges manage enquiries through manual registers, phone calls, or disjointed email systems. This project digitizes that process by providing a centralized platform for prospective students to explore courses, view faculty details, and submit admission queries online.

1.2 Problem Statement

Manual admission enquiry systems are prone to human errors, data loss, and delays in communication. Prospective students often struggle to find consolidated information about courses, fees, and eligibility criteria. Furthermore, the college administration lacks a centralized dashboard to track, filter, and respond to the influx of enquiries efficiently, leading to potential loss of prospective enrollments.

1.3 Objectives of the Project

1. To develop a user-friendly public portal for students to browse courses, faculty, and admission procedures.
2. To implement a secure online enquiry form with proper validation.
3. To create a secure admin dashboard for college staff to view, filter, and update the status of submitted enquiries.
4. To migrate the existing legacy system to a modern Python Flask architecture for improved performance and maintainability.

1.4 Scope of the Project

The scope of this project encompasses the development of a responsive web application featuring two main modules: a Public Facing Portal (Home, About, Courses, Admissions, Faculty, Contact) and a Secure Admin Panel. The system handles data entry, storage, retrieval, and status management for enquiries. It does not currently include online fee payment gateways or a full student lifecycle management system post-admission.

1.5 Significance of the Project

This project is significant as it modernizes the first point of contact between prospective students and the college. By moving from a manual to a digital system, it ensures data integrity, speeds up response times, and presents a professional image of the institution. It also provides valuable data insights to the administration regarding popular courses and enquiry trends.

Chapter 2 - Literature Review / Background Study

2.1 Introduction to the Subject Area

The subject area of this project lies at the intersection of Web Application Development and Educational Management Information Systems (EMIS). Web-based systems have become the standard for organizational management due to their accessibility and cross-platform compatibility.

2.2 Existing Systems / Related Work

Many institutions currently employ standalone Student Information Systems (SIS) or generic Customer Relationship Management (CRM) software. While powerful, these systems are often overly complex, expensive, and require significant training to operate. Alternatively, some colleges rely on simple Google Forms or physical paper trails, which lack proper tracking and status management capabilities.

2.3 Limitations of Existing Systems

1. Manual registers are difficult to search and prone to physical damage or loss.
2. Email-based systems lack a structured way to track the status of an enquiry (e.g., Pending, Contacted, Admitted).
3. Enterprise CRM systems are often too expensive and complex for small to medium-sized colleges.
4. Lack of integration between the public-facing website and the backend data management system.

Chapter 3 - System Analysis and Requirements

3.1 Existing System

The existing system for managing admission enquiries at many similar institutions is largely manual. Students visit the campus physically or make phone calls to inquire about courses. Staff members note down details in physical registers. Tracking follow-ups is tedious, and there is no centralized database to analyze enquiry trends.

3.2 Proposed System

The proposed system is a digital College Admission Enquiry System. It provides a public website where students can find information and fill out an enquiry form. The submitted data is instantly stored in a secure MySQL database. College administrators can log in to a backend dashboard to view these enquiries, filter them by course or date, and update their status as the admission process progresses.

3.3 Functional Requirements

1. Public Portal: Users must be able to view Home, About, Courses, Admissions, Faculty, and Contact pages.
2. Enquiry Submission: Users must be able to submit a form containing their personal details, educational background, and preferred course.
3. Admin Authentication: Administrators must be able to log in using secure credentials.
4. Dashboard Analytics: The admin panel must display summary statistics (total enquiries, pending, contacted, etc.).
5. Enquiry Management: Admins must be able to view full details of an enquiry, update its status, or delete it.

3.4 Non-Functional Requirements

1. Security: Admin routes must be protected. Passwords must be hashed. Inputs must be sanitized to prevent SQL injection.
2. Performance: The system should load quickly and handle multiple concurrent users.
3. Usability: The interface must be intuitive, responsive, and accessible on mobile devices.
4. Reliability: The database must ensure data integrity and persistence.

3.5 Hardware Requirements

Server						Side:
-	Processor:	Intel	Core	i3	or equivalent	(minimum)
-		RAM:			4GB	minimum
-		Storage:			20GB	SSD/HDD

Client		Side:
- Any device (PC, Tablet, Smartphone) with an internet connection.		

3.6 Software Requirements

Server						Side:
-	Operating		System:		Windows/Linux/macOS	
-	Programming		Language:		Python	3.8+
-	Web		Framework:		Flask	
-			Database:		MySQL	
-	Web	Server:	Werkzeug	(development)	/	Gunicorn (production)

Client		Side:
- Web Browser: Google Chrome, Mozilla Firefox, Safari, or Microsoft Edge.		

Chapter 4 - System Design and Methodology

4.1 Development Methodology

The Agile software development methodology was adopted for this project. This iterative approach allowed for continuous feedback and incremental improvements. The project was broken down into phases: Planning, Design, Backend Development, Frontend Integration, and Testing.

4.2 System Architecture

The system follows a Model-View-Controller (MVC) architectural pattern, adapted for the Flask framework.

- Model: Handles database connections and data logic (MySQL queries).
- View: Renders the user interface using Jinja2 templates, HTML, and CSS.
- Controller: Flask routes in app.py that process incoming requests, interact with the model, and return the appropriate view.

4.3 Database Schema

The database consists of the following primary tables:

1. admins: Stores admin credentials (id, username, password hash).
2. admission_enquiries: Stores submitted enquiries (id, full_name, email, phone, course, status, created_at, etc.).
3. courses: Stores course details (id, course_name, short_name, duration, fee, eligibility).
4. faculty: Stores faculty information (id, name, designation, department, qualification).
5. subjects: Stores semester-wise subject details mapped to courses.
6. faqs: Stores frequently asked questions.

Chapter 5 - Implementation

5.1 Technologies Used

1. Python: The core programming language used for backend logic.
2. Flask: A lightweight WSGI web application framework for routing and handling HTTP requests.
3. PyMySQL: A pure-Python MySQL client library used for database interactions.
4. HTML5 & CSS3: Used for structuring and styling the web pages.
5. Jinja2: The templating engine used to dynamically render HTML pages with data from Python.
6. FontAwesome: Used for scalable vector icons.

5.2 Major Modules

1. Public Website Module: Renders static information and dynamic content from the database (courses, faculty).
2. Form Handling Module: Validates incoming form data on both the client-side (HTML5) and server-side (Python/Regex) before inserting it into the database.
3. Admin Authentication Module: Manages session creation, password hashing (werkzeug.security), and route protection (@login_required decorator).
4. Admin Dashboard Module: Aggregates data for statistical display and provides CRUD (Create, Read, Update, Delete) interfaces for managing enquiries.

Chapter 6 - Testing and Results

6.1 Testing Methodology

Manual testing was performed on the application to ensure all functionalities work as expected. This included Unit Testing for individual functions (e.g., email validation), Integration Testing for database connections, and User Acceptance Testing (UAT) for the overall user interface flow.

6.2 Test Cases

The following test cases demonstrate the validation and functionality checks performed on the system:

Test Case	Input	Expected Result	Actual Result	Status
Admin Login	Valid username & password	Redirect to Dashboard	Redirected to Dashboard	Pass
Admin Login	Invalid password	Show error message	Error message displayed	Pass
Access Admin Area	Unauthenticated access attempt	Redirect to Login page	Redirected to Login	Pass
Submit Enquiry	All valid data	Save to DB, show success msg	Saved, success msg shown	Pass
Submit Enquiry	Invalid email format	Show validation error	Validation error shown	Pass
Submit Enquiry	Phone < 10 digits	Show validation error	Validation error shown	Pass
Update Status	Change to "Contacted"	Status updated in DB	Status updated successfully	Pass
Delete Enquiry	Click delete on enquiry	Record removed from DB	Record removed successfully	Pass

Chapter 7 - Conclusion and Future Scope

7.1 Conclusion

The College Admission Enquiry System has been successfully developed and implemented using the Python Flask framework. The project meets all its primary objectives, providing a seamless digital experience for prospective students to submit enquiries and a robust, secure backend for administrators to manage them. The migration from a legacy PHP architecture to a modern Python stack has resulted in cleaner, more maintainable code.

7.2 Limitations

Currently, the system is focused solely on enquiry management. It does not handle the complete admission workflow, such as document uploads, online fee collection, or automated email notifications to students upon status updates.

7.3 Future Scope

In the future, the system can be expanded to include:

1. Integration with a payment gateway for online admission fee collection.
2. Automated email and SMS notifications triggered when an enquiry status changes.
3. A full student portal for tracking admission progress and uploading necessary verification documents.
4. An AI-powered chatbot backend to handle common student queries automatically.